
EE 601: Advanced Electric Drives

Course Code	EE 601
Course Name	Advanced Electric Drives
L-T-P-C	2.5-0.5-0-3
Prerequisites	EE 201 Electromechanics and Power Electronics
Intended for	B.Tech. and MS/Ph.D.
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Introduction:** Definition of electric drive, type of drives; Speed torque characteristic of driven unit/loads, motors, joint speed-torque characteristic; Classification and components of load torque; Review of power converters used in drives, multi-quadrant operation of electric drive, example of hoist operation in four quadrant.
- **Module II:** Closed loop control of solid state DC drives, Scalar and vector control of induction motor, Direct torque and flux control of induction motor, Self controlled synchronous motor drive, Vector control of synchronous motor, Switched reluctance motor drive, Brushless DC motor drive, Permanent magnet drives, Industrial drives.
- **Module III:** Harmonic reduction techniques, PWM inverters, Space Vector Modulation

Text and Reference Books

1. Mohan N., Underland T.M. and Robbins W.P., **Power Electronics Converters, Applications and Design**, 3rd Edition, Wiley India. 2008
2. Bose B.K., **Power Electronics and Variable Frequency Drives Technology and Applications**, IEEE Press, Standard Publisher Distributors. 2001
3. B.K.Bose, **Power Electronics & A.C. Drives**, Prentice Hall, 1986.
4. Rashid M., **Power Electronics- Circuits, Devices and Applications**, 3rd Edition, Pearson Education.
5. Dubey G. K., **Power Semiconductor Controlled Drives**, Prentice Hall, 1989
6. Murphy J. M. D. and Turnbull F. G., **Power Electronics Control of AC Motors**, Peragmon Press.
7. G.K.Dubey, **Fundamentals of Electric Drives**.